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# Trading threshold

*by John Ehlers*

In radar, signal-to-noise ratio is used to measure the quality of target detection. In trading, this simple concept can be used to hit our profit targets better. We can improve our trading profitability if trading decisions are deferred until the signal-to-noise ratio is high. The charts can be viewed as noisy channels in which the daily range and very short-term day-to-day variations are the "noise." The longer-term variations mark the channel envelope.

The trader's profitability can be enhanced if he waits for conditions in which the peak-to-peak variation of the channel envelope exceeds four times the width of the "noise band." Noise as classically defined is energy (in our case, price activity) that carries no information. Complete randomness has no information and, therefore, random events are noisy. When noise is completely random, all frequencies are present. This is called "white noise" because the picture of the uniform frequency distribution resembles snow.

White noise has a Gaussian, or normal, amplitude probability distribution, which is the familiar bell-shaped curve that describes many statistical cases, such as a given population's IQ distribution. In the same way very few people have exceptionally high IQs, the amplitude of white noise can be very large but occur very seldom.

The power of signals and noise are most conveniently measured in decibels (dB). "Deci-" means one-tenth, while "bel" is a logarithmic of the ratio of two power values. Since the logarithm of 2 is 0.3, each doubling of the power ratio results in a 3 dB increase ($\log_{10}2 = 0.3$, so $lO\log_{10}2= 3$). For example, 6 dB is the value of a power ratio of 4 ($\log_{10}4 \approx 0.6$ or $10^{0.6} = 4$). The logarithm of 10 is 1, so the power ratio of 10 is also expressed as 10 dB. Power ratios can be halved as well as doubled. For example, 7 dB is 3 dB less than a power ratio of 10, and thus, the 7 dB power ratio is half, or 5-to-1.

Note how radar signals are detected. Figure 1 shows a simulated radar pulse imbedded in white noise

  

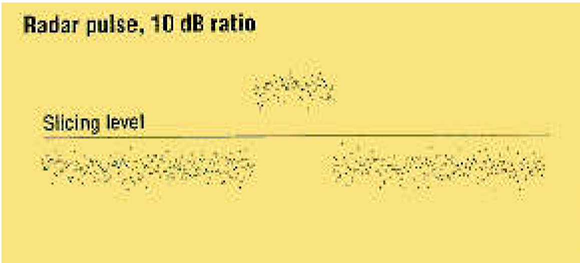


**FIGURE 1:** *A radar pulse with a 10 dB signal-to-noise ratio is easily detected by the sharp separation between noise and the pulse.*

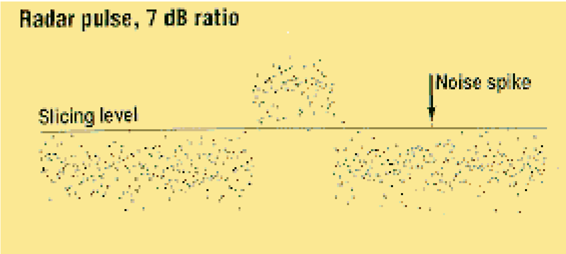


**FIGURE 2:** *A radar pulse with a 7 dB signal-to-noise ratio is not as sharply defined as a 10 dB pulse so the slicing level must be more carefully chosen.*

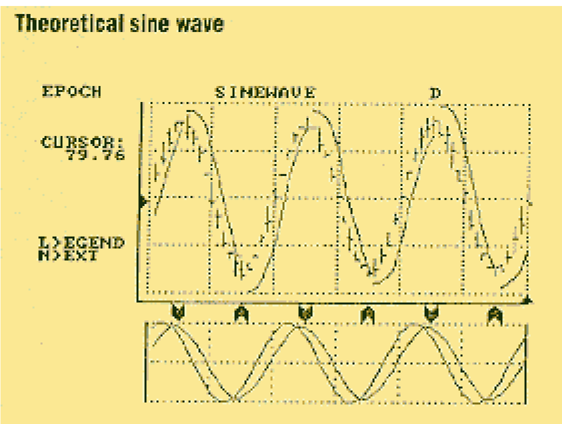


**FIGURE 3:** *Trading signals for a theoretical sine wave with no noise are virtually perfect*.

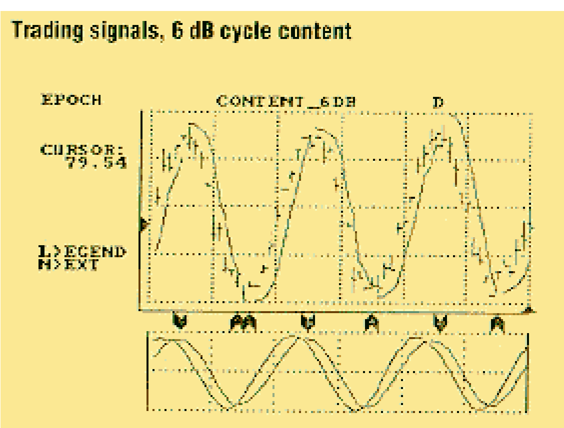


**FIGURE 4:** *Trading signals with a 6 dB signal-to-noise ratio are still fairly good*.

with a 10 dB signal-to-noise ratio. To detect this pulse, simply establish a "slicing level" so the signal plus noise is always above this level and noise alone is always below it.

The problem becomes more severe when the signal-to-noise ratio is reduced to 7 dB, as shown in Figure 2. The 7 dB signal-to-noise case is commonly called the "tangential sensitivity" of the radar. When the slicing level is set, the presence of the pulse can be correctly identified most of the time. The most common errors are classifying a noise spike as a pulse or missing the pulse identification when the noise of the pulse falls below the given threshold. The ability to properly identify the pulse deteriorates rapidly as the signal-to-noise ratio falls below 7 dB.

The decision to trade is also influenced by the signal-to-noise ratio. In my cycle-based trading programs this ratio is referred to as "cycle content." A noise-free trading example using a theoretical sine wave is shown in Figure 3. The trade entry points are indicated at the bottom of the graph, and the stop is the smooth line discontinuous only with the trade reversal. The trade is not severely hampered when the cycle content is reduced to 6 dB, as shown in Figure 4. But when cycle content is zero (when noise power is equal to the cycle power), noise whipsaws the trades for losses as in Figure 5.

Why is a 6 dB cycle content useful in trading when a 7 dB signal-to-noise ratio is required for good radar pulse detection? Not all noise is white noise with a Gaussian distribution. Intraday and daily trading limits place restrictions on the peak excursions for noise spikes, which make the probability distributions more like a uniform probability between two bounds. When uniform probability noise is superimposed on a simulated radar pulse with a 6 dB signal-to-noise ratio, we have the result shown in Figure 6. Note the edges of the noise are less fuzzy, allowing tangential sensitivity to be about 1 dB lower than it would be with white noise.

A sine wave with uniform probability density noise super- imposed at a 6 dB signal-to-noise ratio is shown in Figure 7. The sine wave can be viewed as a trading channel or cycle.

The signal-to-noise ratios of real signals can be estimated using Figure 7 as a guide. We know the power ratio is 4-to- 1 ($\log_{10} 4 \approx 0.6$) — that is, the signal strength is four times the strength of the noise. Take the noise strength as the width of the channel, preferably near a minimum or maximum signal, and that is the standard amplitude. Next, measure the signal plus noise amplitude from the approximate lowest low to the approximate highest high. The signal amplitude is this measurement, less the channel width, because we have half a channel width extra excursion at both the peak and the valley. Therefore, the ratio of the signal amplitude (peak-to-peak less channel width) divided by the channel width is our signal-to-noise ratio.

> **Why is a 6 dB cycle content useful in trading when a 7 dB signal-to-noise ratio is required for good radar pulse detection?**

The trading rule is simple using the following threshold: Don't trade when the signal-to-noise ratio is below 6 dB (a 4-to-l ratio).

### Basic programming

You can experiment with the impact of various signal-to-noise ratios using the BASIC computer program in Figure 8. The program is written in generic syntax but requires a CGA monitor to observe the graphic

  

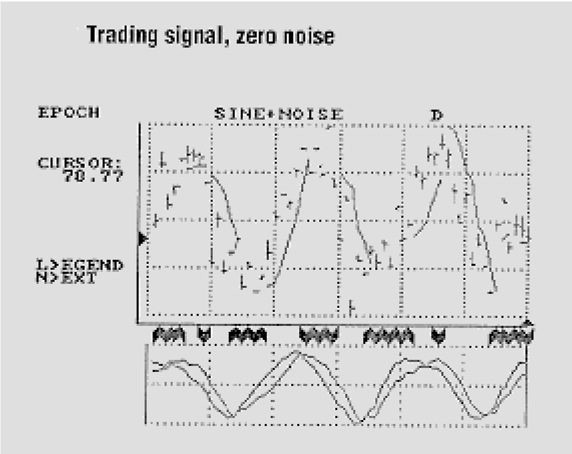


**FIGURE 5:** *Trading signals without noise are confused and unprofitable.*

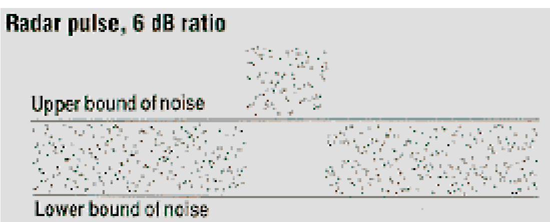


**FIGURE 6:** *A radar pulse with a 6 dB signal-to-noise ratio — a uniform, bounded probability of noise distribution — allows identification of a radar pulse.*


**FIGURE 7:** *A 6 dB signal-to-noise trading channel is a bounded, uniform distribution.*

**Signal-to-noise BASIC program**

```
10 REM *** WRITTEN BY JOHN EHLERS ***
20 SCREEN 2
30 DIM SIG(700), NOISE(700)
40 REM *** REPEATS FOR NEW SELECTION OF SNR ***
50 CLS
60 INPUT "SIGNAL TO NOISE RATIO (IN DB)"; SNR
70 SNR = 10 ^ (SNR / 30)
80 FOR I = 0 TO 270
90 SIG(I) = -.5
100 NEXT
110 FOR I = 271 TO 370
120 SIG(I) = .5
130 NEXT
140 FOR I = 371 TO 639
150 SIG(I) = -.5
160 NEXT
170 FOR I = 1 TO 639
180 PSET (I, 100 - 50 * (SIG(I) + 3.46 * (RND - .5) / SNR))
190 NEXT
200 S$ = INKEY$
210 IF S$ <> "" THEN GOTO 200
220 S$ = INKEY$
230 IF S$ = "" THEN GOTO 220
240 CLS
250 FOR I = 1 TO 639
260 PSET (I, 100 - 30 * (1.414 * SIN(6.28 * I / 200) + 3.46 * (RND - .5) / SNR))
270 NEXT
280 S$ = INKEY$
290 IF S$ <> "" THEN GOTO 280
300 S$ = INKEY$
310 IF S$ = "" THEN GOTO 300
320 CLS
330 FOR I = 0 TO 639
340 NOISE(I) = 0
350 FOR J = 1 TO 100
360 NOISE(I) = NOISE(I) + (RND - .5)
370 NEXT
380 NOISE(I) = NOISE(I) / 10
390 NEXT
400 FOR I = 1 TO 639
410 PSET (I, 100 - 50 * (SIG(I) + 3.46 * NOISE(I) / SNR))
420 NEXT
430 S$ = INKEY$
440 IF S$ <> "" THEN GOTO 430
450 S$ = INKEY$
460 IF S$ = "" THEN GOTO 450
470 CLS
480 FOR I = 1 TO 639
490 PSET (I, 100 - 30 * (1.414 * SIN(6.28 * I / 200) + 3.46 * NOISE(I) / SNR))
500 NEXT
510 S$ = ""
520 IF S$ <> "" THEN GOTO 510
530 S$ = INKEY$
540 IF S$ = "Q" OR S$ = "q" THEN END
550 IF S$ = "" THEN GOTO 530
560 GOTO 40
```

**FIGURE 8 :** *You can experiment with the impact of various signal-to-noise ratios using the BASIC computer program.*

**Example of a noisy but tradeable trading channel**

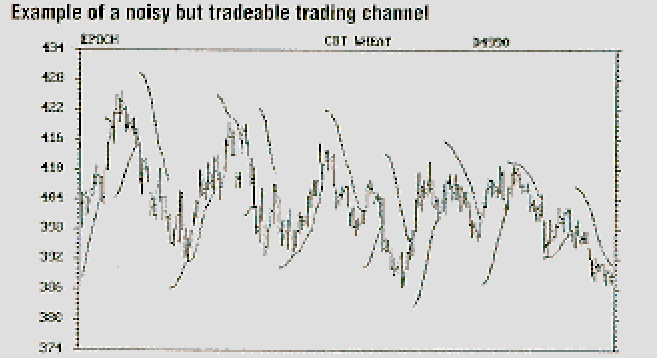


**FIGURE 9:** *These data contain a fair amount of noise but remain tradeable, nevertheless.*

response.

Lines 80 to 190 calculate and display signal-plus-noise for a simulated radar pulse for uniform probability noise. The BASIC random number generator is the noise source. The generator outputs a random number between zero and 1. This random number is offset by -0.5 so the noise has a zero mean value. Lines 250 to 270 calculate and display the shape of a sine wave plus the uniform probability noise.

Invoked in lines 330 through 390 is the central limit theorem, which states that the (large) sum of random events will produce a Gaussian probability distribution of the resulting random function. The sum is normalized to a constant reference amplitude by the square root of the number of events summed in line 380.

The approximated white noise is added to a radar pulse in lines 400 to 420. White noise is also added to the sine wave and displayed in lines 480 to 500. The program will terminate if you press "Q" for quit after display of the fourth screen. Pressing any other key will take you back to the beginning of the program. Computing the Gaussian noise can take quite a while, particularly if you are using interpreted BASIC on an older personal computer. Patience!

The "thresholding" logic for the detection of a radar pulse in the presence of noise generates a simple rule for knowing when to trade. The rule is to avoid trading when the signal-to-noise ratio is less than 6 dB. The 6 dB threshold is measured as the peak-to-peak variation less the noise channel width, divided by the noise channel width.

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**Reference**

Ehlers, John [1989]. "Cyclic personalities," *Technical Analysis of Stocks & Commodities Volume 7*.